

THE QUESTION

THE INTUITION

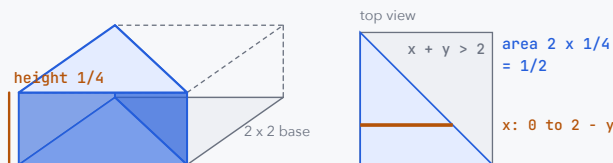
KEY TAKEAWAY

joint $p(x, y)$ $\rightarrow$ add a row: $p(x)$ $\rightarrow$ add a column: $p(y)$ $\rightarrow$ row / its total: $p(y | x)$ $\rightarrow$ product test $\rightarrow$ density: volume

Joint densities: probability is volume

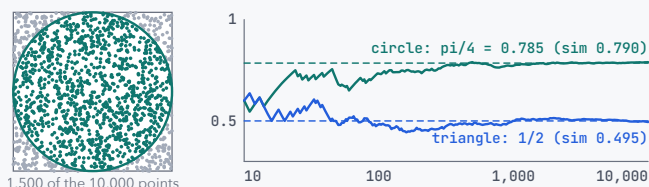
For measured pairs, integrals replace sums, and a region's probability is the volume above it.

7 Probability is the volume under $f(x, y)$



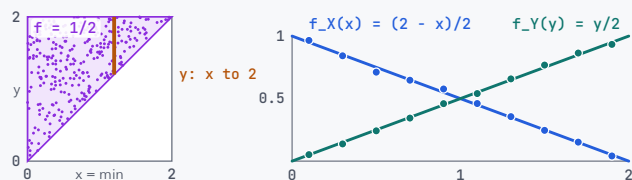
$P((X, Y) \in A) = \iint_A f(x, y) dx dy$, the volume above A . $f \geq 0$ with total volume 1; f can exceed 1, and a line has probability 0. Uniform on $[0, 2]^2$: $f = 1/4$, one over the base area. For $x + y \leq 2$: $\int_0^2 \int_0^{2-y} 1/4 dx dy = \int_0^2 (2-y)/4 dy = 1/2$.

8 A dart lands in the inscribed circle with chance $\pi/4$



With a flat density a region's chance is its area over 4. The circle that just fits has area π : chance $\pi/4 = 78.5\%$. Of 10,000 uniform points (seed 0), 0.790 land inside ($\pi \approx 3.16$) and 0.495 below the diagonal; the error shrinks like $1/\sqrt{n}$ (the sums, LLN and CLT sheet).

9 Integrate one variable out to get a marginal



$f_X(x) = \int f(x, y) dy$ adds a row, continuously. The min and max of two Uniform(0, 2) draws fill the triangle $0 \leq x \leq y \leq 2$ with $f = 1/2$, one over its area. So $f_X(x) = \int_x^2 1/2 dy = (2-x)/2$ and $f_Y(y) = y/2$, the dice's 11, ..., 1 and 1, ..., 11 as densities (dots: 20,000 draws). $f(y | x) = 1/(2-x)$ is flat on $[x, 2]$.

12 Pause and think

X, Y each 0 or 1 with chance 1/2. Can $P(X = Y)$ be 1? Yes: 0.5 on (0, 0) and (1, 1).

A constant density on a disc of radius 1: its height? $1/\pi \approx 0.318$, one over the area.

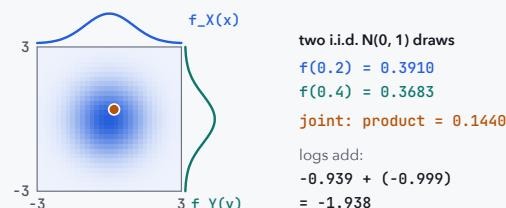
Independent Bernoulli(0.3) and Bernoulli(0.5): $P(X + Y = 1)$? $0.3 \cdot 0.5 + 0.7 \cdot 0.5 = 0.5$.

Every cell is a product. What is $p(y | x)$? $p_Y(y)$: knowing x changes nothing.

Uniform on $[0, 2]^2$: $P(X < 1, Y < 1)$, $P(X = Y)$? $1/4$ (simulated 0.242); 0, a line has no area.

Independent implies uncorrelated. The reverse? No: $Y = X^2$ (the covariance and correlation sheet).

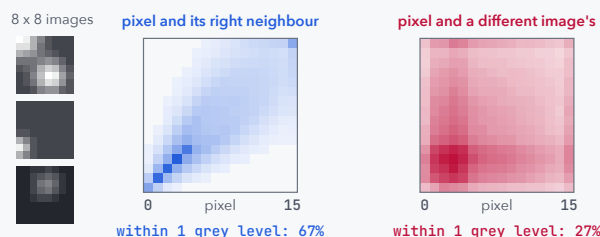
10 Independent densities multiply, and so do i.i.d. draws



Independence: $f(x, y) = f_X(x)f_Y(y)$ for all x, y . The square's $1/4 = 1/2 \times 1/2$; the triangle's support is not a rectangle, so its min and max are dependent. n i.i.d. draws have joint density $\prod_i f(x_i)$ and log density $\sum_i \log f(x_i)$. For 1,000 normal draws (seed 0) the log is -1397.1 while the product underflows to 0.0, one reason libraries return $\log_{\text{prob}}$.

```
x = np.array([0.2, 0.4]) # i.i.d. N(0, 1)
logf = -x**2 / 2 - 0.5 * np.log(2 * np.pi)
s = logf.sum()
s, np.exp(s) # -1.938, 0.144
```

11 A 2D histogram shows that nearby pixels depend



An 8×8 grey image is one draw of 64 jointly distributed pixels. In 2,000 made-up blob images (grey 0 to 15, seed 0), a 2D histogram of each pixel and its right neighbour is a diagonal band: 67% of pairs are within 1 level. A pixel and the same spot in another image: 27%, near the 26% the product of the margins predicts. Pixels drawn from their own marginals would give speckle, not blobs.

$$p(x) = \sum_y p(x, y) \cdot p(y | x) = p(x, y) / p(y) \cdot \text{independent: } p(x, y) = p(x) p(y) \text{ in every cell} \cdot P(A) = \text{volume over } A$$

Source: the course notebooks "Distributions in 2D", "Independent and identically distributed (i.i.d)" (two $\mathcal{N}(0, 1)$ draws at 0.2 and 0.4) and "Images (joint distribution)", and the slides "Integration of joint PDF in a region" (uniform on $[0, 2]^2$, $P(X + Y \leq 2) = 1/2$), nipunbatra.github.io/psdv-teaching. The shop table, dice, triangle and blob images are our own; all numbers come from the sheet's figure script (seed 0). Next: the covariance, correlation and bivariate normal sheet.